

First Sessional Test –September, 2025
B.Tech(Mechanical Engineering)- I-SEMESTER
MATHEMATICS - I(Calculus and Linear Algebra) (BSC-103A)

Time: 90 Minutes

Max. Marks: 15

Instructions: Attempt any three questions and each question carries equal marks:

Q.1. Solve that the system of equations: [CO5]
 $3x + 4y + 5z = a, 4x + 5y + 6z = b, 5x + 6y + 7z = c,$
does not have a solution unless $a + c = 2b$.

Q.2. Let λ be an eigenvalue of a non-singular matrix A and X [CO5]
be the corresponding eigenvector. Then, $\text{adj}(A)$ has the eigenvalue
 $\frac{|A|}{\lambda}$ with the eigenvector X .

Q.3. If $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, find A^n . [CO5]

Q.4. If $[x]$ denotes the greatest integer less than or equal to x , then, evaluate [CO2]

$$\lim_{n \rightarrow \infty} \frac{1}{n} ([1^2 x] + [2^2 x] + [3^2 x] + \dots + [n^2 x]).$$

$$\begin{aligned} & \left(c - \frac{5}{3}a\right) - 2\left(b - \frac{4}{3}a\right) \\ & c - 2b - \frac{5}{3}a + \frac{8}{3}a \end{aligned}$$